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Bayesian Linear Regression Simulation & Analysis

Introduction

Life expectancy varies dramatically between countries, reflecting differences in access to technology, education, and economic resources. Understanding how these factors shape health outcomes is critical for informing public policy and guiding international development initiatives. Prior research shows that higher internet access, better education, and stronger economic development are associated with longer life expectancy, yet traditional regression approaches provide only point estimates and often fail to fully quantify uncertainty.

The goal of this honors contract analysis is to estimate the effects of digital access, education, and GDP on life expectancy using a Bayesian linear regression framework, which combines prior knowledge with observed data to produce posterior distributions that capture both the size and uncertainty of these effects. The central question is: *how uncertain are the effects of digital access, education, and GDP on life expectancy across countries?*

Data and Preprocessing

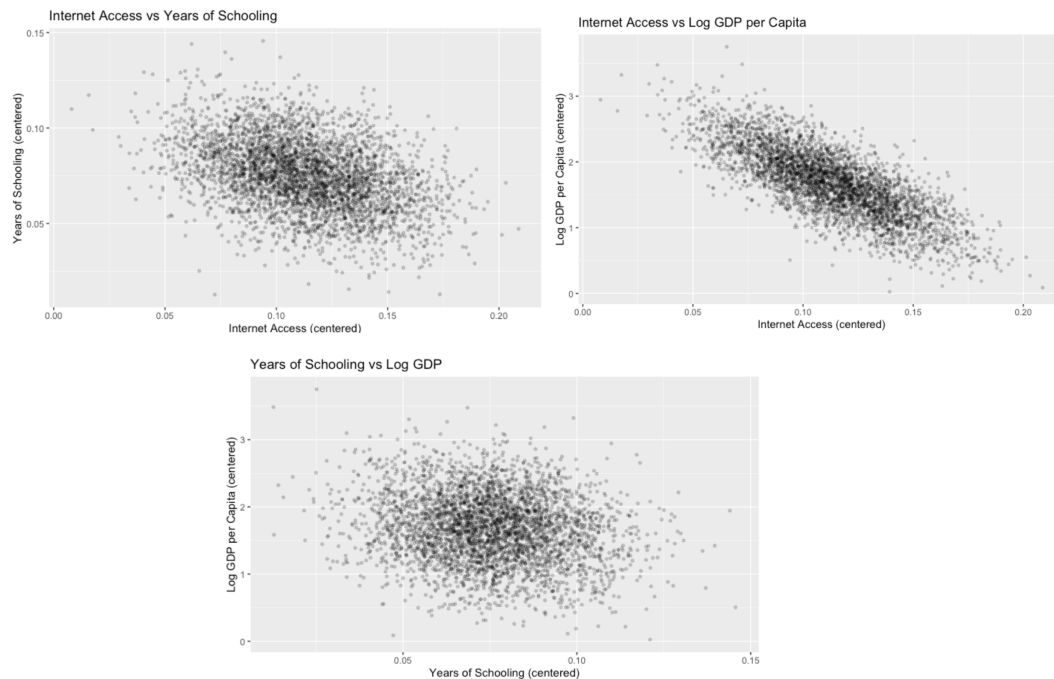
Data were drawn from the World Bank, an international financial institution that collects and publishes country-level indicators on health, education, and economic outcomes, for the years 2015–2020. The analysis focuses on country-level averages of the following variables:

- **Internet users (% of population):** proxy for digital access
- **Secondary school enrollment (% gross):** proxy for educational attainment
- **GDP per capita (USD):** a measure of economic development
- **Life expectancy at birth (years):** outcome variable of interest

For this analysis, only countries with complete data on all four variables were included. While this approach ensures valid modeling, it may introduce bias if countries with missing data systematically differ in life expectancy or development indicators. Country-level averages were calculated across 2015–2020 to smooth year-to-year variation. GDP was log-transformed to reduce skewness because GDP tends to be highly skewed due to a small number of very wealthy countries; this transformation stabilizes variance and improves the validity of regression assumptions, including linearity and homoscedasticity. All predictors were mean-centered, which improves the interpretability of the intercept, representing an “average” country. This also ensures that prior distributions on the slope coefficients are reasonable over the observed range, avoiding unrealistic prior implications.

Posterior Collinearity Diagnostics

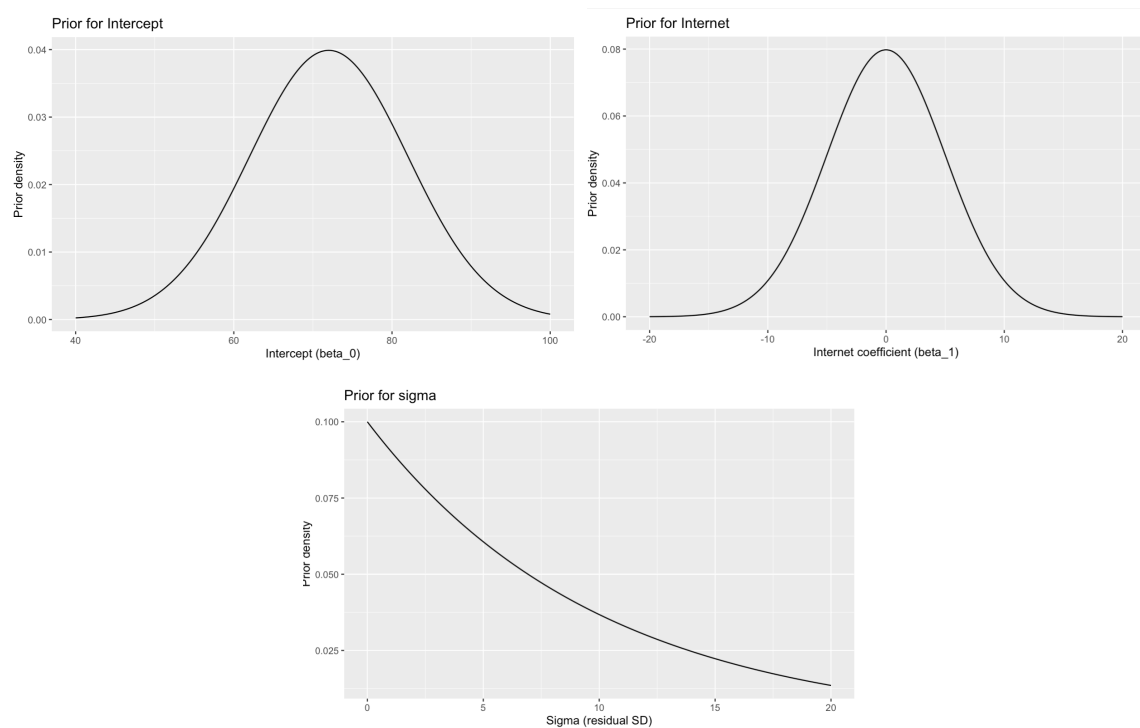
To evaluate potential collinearity among predictors in the Bayesian regression model, I examined scatter plots of the joint posterior samples for each pair of regression coefficients. These plots visualize how strongly the posterior draws for two coefficients move together, which provides a direct diagnostic of collinearity from the model's perspective. The scatter plot comparing the Internet Access and Log GDP per Capita coefficients shows a negative diagonal pattern, suggesting collinearity between these two predictors. In contrast, the relationship between Internet Access and Years of Schooling displays only a mild elliptical shape, suggesting slight collinearity. The scatter plot for Years of Schooling and Log GDP shows a less structured, circular cloud of points, suggesting very little collinearity between these predictors.



Prior Specification

Bayesian modeling allows explicit incorporation of prior beliefs, guiding estimation and quantifying uncertainty when data are limited or noisy. For this analysis, the intercept prior was specified as $\text{Normal}(72, 10)$, centered at the global mean life expectancy. This choice reflects the expectation that a “typical” country with average digital access, education, and economic development would have a life expectancy near the global mean, while the standard deviation of 10 accommodates substantial variation across countries (roughly 50–85 years) without allowing extreme values. The slope priors for internet access, school enrollment, and $\log(\text{GDP})$ were specified as $\text{Normal}(0, 5)$, centered at zero to reflect prior ignorance about the direction of the effects. The standard deviation of 5 is large enough to allow meaningful positive or negative effects on life expectancy, yet small enough to exclude unrealistic magnitudes that would imply, for example, an implausibly large increase in life expectancy from a one-unit change in a

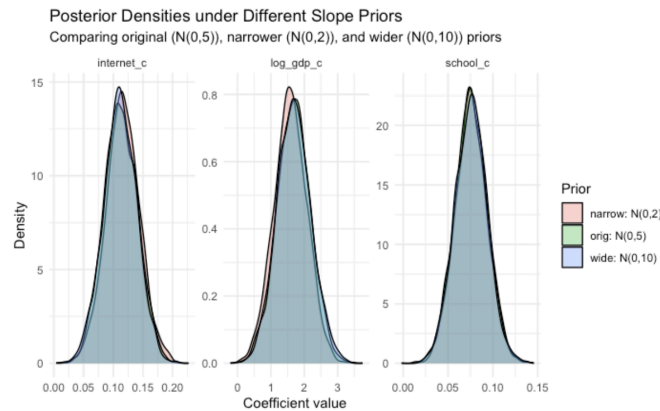
predictor. Finally, the residual standard deviation was given an Exponential(rate = 0.1) prior. This choice reflects typical variability in life expectancy not explained by the predictors and ensures that the residual SD remains positive, as required for a valid model. Prior predictive plots confirm that these distributions produce plausible ranges for the intercept, slopes, and residual variance, providing a foundation for combining prior beliefs with observed data to estimate the posterior distributions.



Prior Sensitivity Analysis

To evaluate how conclusions depend on prior choices, the model was refitted with two alternative slope priors: a more concentrated prior, $\text{Normal}(0,2)$, and a more loose, broad prior, $\text{Normal}(0,10)$. All other model settings were unchanged. The figure below shows the overlaid posterior density estimates for the three slope coefficients under each prior specification, which can allow us to assess how strongly the Bayesian regression results depend on the choice of prior. The curves for all three priors (narrow, original, and wide) almost perfectly overlap, indicating that the posterior distributions for internet access, log GDP per capita, and schooling remain nearly identical regardless of the prior use. Although the narrower prior produces slightly more concentrated distributions, which is especially distinct in the log GDP coefficient, the differences across priors are very small. In the same vein, the wide prior produces only minimal

broadening. The near-complete overlap of the curves indicates that the likelihood dominates the prior for all three predictors, and the data provide strong information about these coefficients.



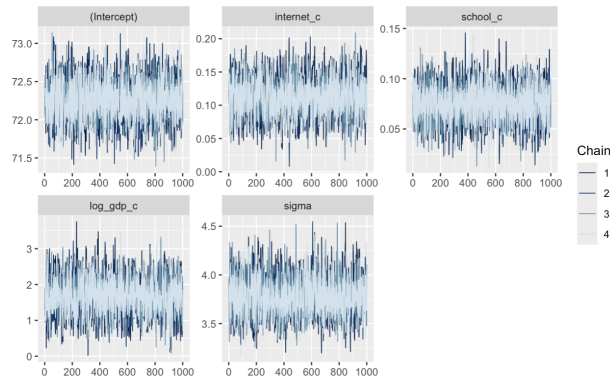
Bayesian Linear Regression Model

To quantify the relationships between digital access, education, GDP, and life expectancy, I fit a Bayesian linear regression model:

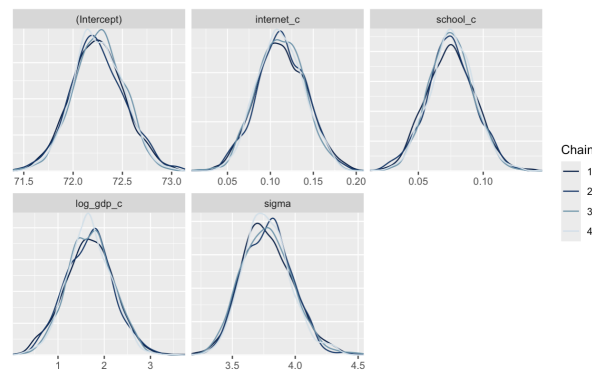
$$\text{Life expectancy} \sim \text{internet use (centered)} + \text{school enrollment (centered)} + \log(\text{GDP per capita (centered)})$$

While the assignment requested posterior inference using a simple Monte Carlo simulation, this analysis employed full Markov Chain Monte Carlo sampling to more rigorously approximate the posterior distribution. The model was estimated using *rstanarm*, generating 20,000 posterior draws through MCMC sampling (4 chains, 2,000 iterations each). This approach combines prior beliefs with observed data to approximate the full posterior distribution of model parameters, including the intercept, slopes, and residual standard deviation. Convergence diagnostics were performed to verify the reliability of posterior estimates:

1. Rhat values were approximately 1 for all parameters, indicating that chains converged to the same posterior distribution. Values above 1.1 would suggest potential convergence issues.
2. Neff_ratio values were close to 1, reflecting a high effective sample size and indicating that the posterior draws are nearly independent. Very low values less than 0.1 would signal poor sampling.
3. Trace plots exhibited the characteristic “fuzzy caterpillar” pattern with no visible trends or sticking points, suggesting that chains explored the parameter space effectively.



4. Density overlays confirmed that the posterior distributions from all chains were consistent and largely overlapping, further supporting convergence.



Posterior Summaries and Inference

Posterior distributions provide full probabilistic estimates of model parameters, which quantifies both effect sizes and the uncertainty surrounding them. Table-based summaries were extracted from the posterior using median estimates and corresponding 95% credible intervals. These intervals represent the range within which the true parameter value lies with 95% posterior probability. Key posterior results include:

- **Intercept (β_0):** The posterior median is approximately 72.2, with a 95% credible interval of [71.7, 72.8]. This represents the expected life expectancy for a “typical” country with average levels of internet access, school enrollment, and GDP because predictors were mean-centered.
- **Internet access (β_1):** The posterior median effect is 0.112, with a credible interval of [0.076, 0.149]. This suggests that a one-standard deviation increase in internet access is associated with an increase of about 0.11 years in life expectancy.
- **School enrollment (β_2):** The median effect is 0.075 with a credible interval of [0.052, 0.099]. This indicates a moderate but consistently positive association between secondary education and life expectancy.

- **Log GDP (β_3):** The effect of economic development is substantially larger, with a median estimate of 1.68 and a credible interval of [1.04, 2.33]. This implies that a one-standard deviation increase in log GDP per capita is associated with an increase of roughly 1.7 years in life expectancy.
- **Residual standard deviation (σ):** The posterior median for unexplained country-level variability is approximately 3.77, with uncertainty captured in the interval [3.53, 4.04].

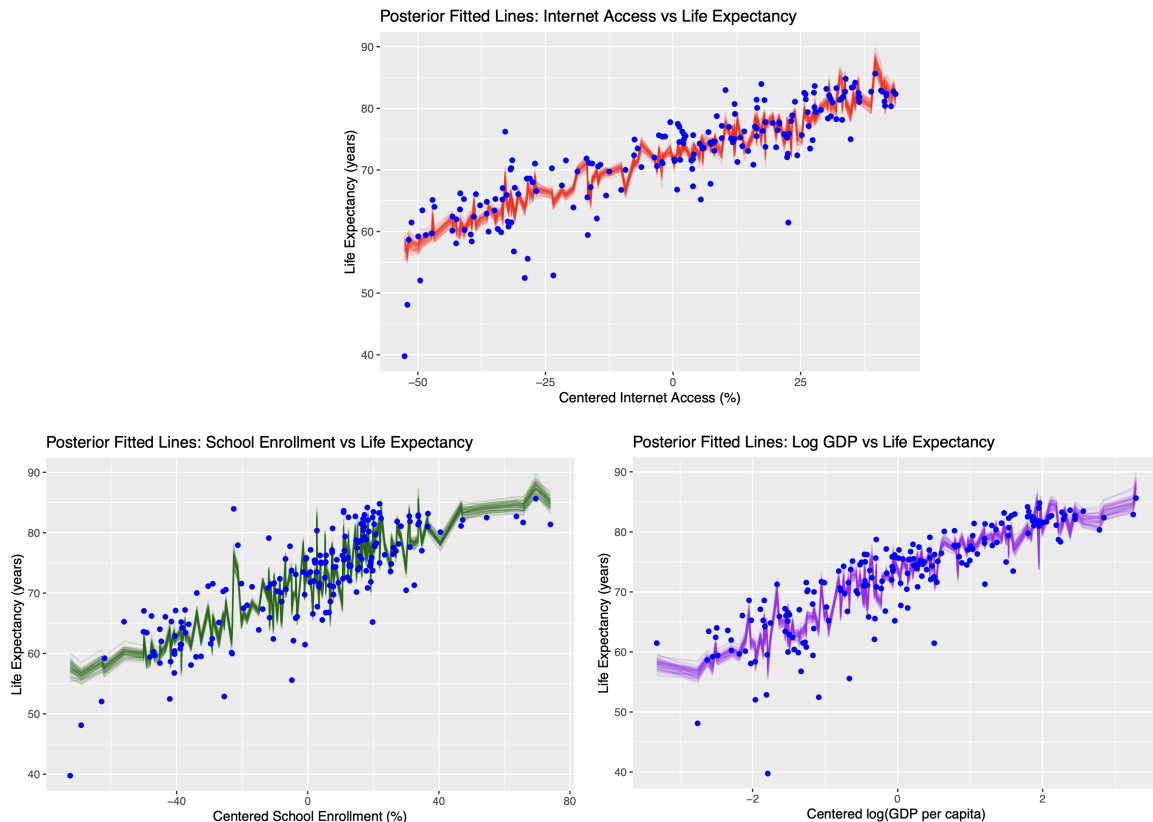
To evaluate the certainty regarding the direction of effects, posterior probabilities were computed for each slope being positive. All predictors had a posterior probability of 1.00 of being greater than zero. This provides strong evidence that internet access, school enrollment, and GDP each have a positive association with life expectancy in the global data. Taken together, these posterior summaries indicate that economic development has the strongest association with life expectancy.

Posterior Fitted Lines

To visualize how the Bayesian model links each predictor to life expectancy, posterior fitted values were generated using `add_fitted_draws()`, which produces draws from the posterior distribution of the fitted regression line. Plotting 50 lines for each predictor provides a clear picture of both the estimated relationship and its associated uncertainty.

Across all three predictors, centered internet access, centered school enrollment, and centered log GDP, the fitted lines show consistent positive slopes. This indicates that, conditional on the other variables in the model, higher digital access, greater school enrollment, and higher GDP are all associated with longer life expectancy. The overlaid lines vary in steepness, reflecting posterior uncertainty in the size of each effect.

The spread of the fitted lines is informative: narrower bands (as seen for internet access and school enrollment) indicate relatively low uncertainty, while wider spreads (especially for log GDP) reflect greater uncertainty about the strength of the economic–health relationship. In each case, observed data points fall close to the cloud of fitted lines, suggesting that the model captures main patterns in the data well and that posterior predictions are consistent with the observed variation.



Posterior Predictive Simulations

To evaluate how the fitted Bayesian model translates predictor values into plausible life expectancy outcomes, posterior predictive simulations were generated for three hypothetical country profiles. These scenarios vary in digital access, educational attainment, and economic development, using centered predictor values:

- **Low development:** Internet = -20, School = -10, log(GDP) = -1
- **Average development:** all predictors = 0
- **High development:** Internet = +20, School = +10, log(GDP) = +1

For each scenario, the model outputs two related quantities, and the results were as follows:

- **Posterior predictions of μ :** the expected life expectancy given the predictors (i.e., the regression mean).
- **Posterior predictive draws of y_{new} :** simulated life expectancy outcomes that include both parameter uncertainty and residual variability.

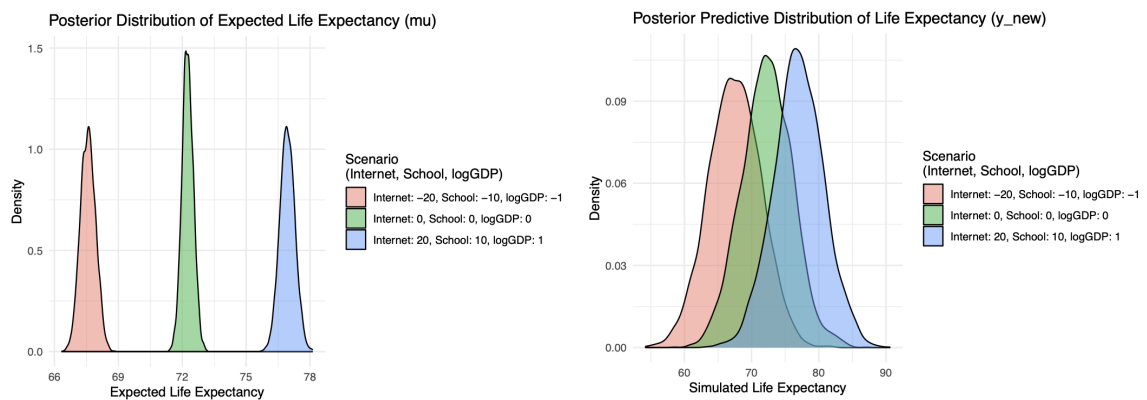
Posterior Expected Life Expectancy (μ)

Scenario	μ	95% CI
Low	67.6	66.9–68.3
Average	72.3	71.7–72.8
High	76.9	76.2–77.6

Posterior Predictive Life Expectancy (y_{new})

Scenario	Mean	95% CI
Low	67.5	59.9–75.0
Average	72.3	64.9–79.7
High	77.0	69.6–84.4

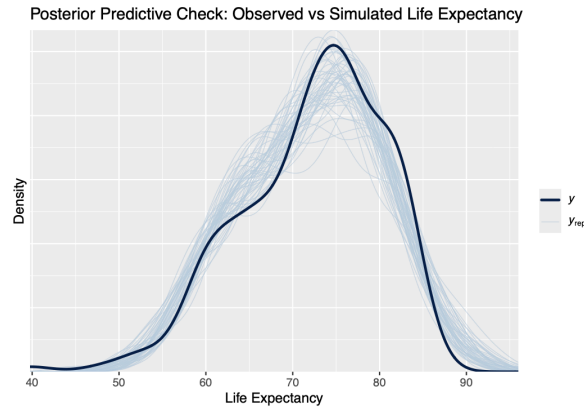
The posterior means show a strong and monotonic relationship between development level and predicted life expectancy. Moving from low to high development shifts μ upward by nearly 10 years, indicating that the model is highly sensitive to changes in digital access, schooling, and economic development. While the μ values differ clearly, the full posterior predictive distributions overlap somewhat. This reflects residual country-level variability ($\sigma \approx 3.8$ years), capturing real-world unpredictability not explained by the three predictors.



Posterior Predictive Checks (PPC)

Posterior predictive checks were conducted to assess how well the Bayesian model reproduces key features of the observed life expectancy data. These checks compare the observed data to samples from the posterior predictive distribution to assess how accurately the model reproduces the observed patterns.

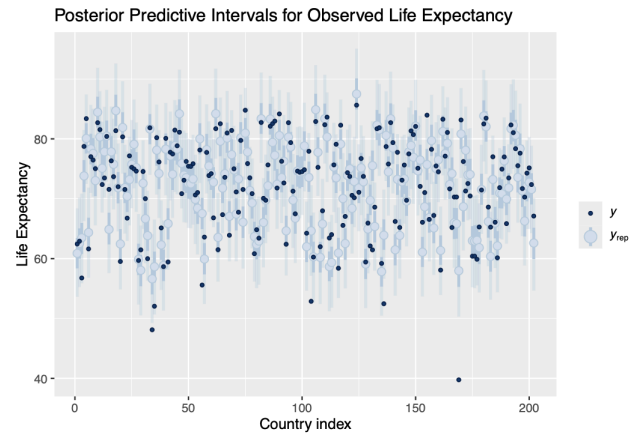
Density Overlay PPC: A density overlay plot compares the observed distribution of life expectancy to 50 replicated datasets generated from the model. The simulated densities closely track the observed distribution, indicating that the model is able to reproduce the overall shape, center, and spread of global life expectancy. This suggests that the model's error structure and assumptions are fit for the data.



Observed vs. Predicted Scatter Plot: The scatter plot of observed versus posterior-predicted life expectancy, with a 45° reference line, shows that most countries fall close to the diagonal. This indicates good pointwise predictive performance. A small deviation appears among countries with very low life expectancy below 55 years, where the model tends to slightly overpredict. This may reflect nonlinearities or unmeasured structural factors in low-income contexts that a simple linear model cannot fully capture.



Posterior Predictive Intervals: The posterior predictive interval plot compares each country's observed life expectancy to its predictive mean and 95% predictive interval. Most observed values fall within their corresponding intervals, consistent with the model's high interval coverage. The few exceptions again occur primarily among the lowest-life-expectancy countries, reflecting limitations in capturing extreme cases. Overall, the intervals illustrate well-calibrated uncertainty: variability in the posterior predictive distribution aligns with variability in the actual data.



Numerical PPC Summary: The quantitative PPC diagnostics confirm these graphical results as shown below:

Metric	Value	Interpretation
MAE	~2.17 years	Predictions are typically within ~2 years of the observed life expectancy.
Scaled MAE	~0.57	Prediction errors are small relative to overall global variability.
Within 50% interval	~0.58	Slightly above the nominal 0.50, indicating minor overconfidence in narrow intervals.
Within 95% interval	~0.96	Very close to the nominal 0.95, proving well-calibrated uncertainty.

Posterior predictive checks indicate that the model accurately reproduces the global distribution of life expectancy and provides reliable point predictions for most countries. It also generates realistic uncertainty intervals with appropriate coverage. The model shows minor underperformance for countries with very low life expectancy, likely due to missing variables or nonlinear effects.

Model Evaluation and Fairness

The model relies on World Bank country-level aggregates rather than individual-level data, which can limit its ability to capture within-country inequalities, such as urban-rural

disparities, income differences, or subgroup variation by gender, age, or ethnicity. Country-level aggregation can overstate certainty for highly diverse nations. In addition, missing data and reporting inconsistencies may introduce bias; only countries with complete predictor data were included, which tends to favor higher-income countries with robust statistical reporting, while underrepresenting low-income countries. Although these results point to meaningful links between development factors and life expectancy that could inform public health planning and policy, they should be interpreted with caution. Future model refinements that incorporate nonlinearities, interactions, and richer within-country data would improve accuracy.

The model itself assumes linear and additive relationships and does not account for nonlinear effects and interactions between predictors. These structural limitations likely contribute to observed deviations in low-life-expectancy contexts. Despite these issues, posterior predictive checks show that uncertainty quantification is generally well-calibrated: roughly 95% of observed life expectancies fall within the model's 95% posterior predictive intervals, and approximately 50% fall within the 50% interval. Although these results point to meaningful links between development factors and life expectancy that could inform public health planning and policy, they should be interpreted with caution. Future model refinements that incorporate nonlinearities, interactions, and richer within-country data would improve accuracy.

What is the Most Important Predictor?

Across all analyses, log GDP per capita emerges as the most important predictor of life expectancy. This conclusion follows directly from the posterior summaries: the median effect of log GDP (≈ 1.68 years per standard deviation) is an order of magnitude larger than the effects of internet access (≈ 0.11) or school enrollment (≈ 0.08). Its 95% credible interval [1.04, 2.33] is well above zero and does not overlap with the intervals for the other predictors, suggesting both a strong and statistically well-separated effect. The prior sensitivity analysis further shows that the posterior distribution of the GDP coefficient remains virtually unchanged under narrow, original, and wide priors, meaning the result is driven by the data rather than prior assumptions. This dominance is also visible in the posterior fitted-line plots, where log GDP shows a positive, steep slope. Collectively, these pieces of evidence indicate that economic development has the strongest association with life expectancy in the data.

Conclusion

This analysis demonstrates that digital access, educational attainment, and economic development are all positively associated with life expectancy across countries, with economic development showing the largest effect. By using a Bayesian linear regression framework, I was able to not only estimate these relationships but also quantify the uncertainty surrounding them. Posterior predictive checks and simulations indicate that the model reliably captures global life expectancy patterns, although it slightly overestimates life expectancy in countries with very low values, likely due to unobserved structural factors and the inflexibility of linear additive models.

Nevertheless, the results suggest that improving access to digital technology, expanding educational opportunities, and fostering economic development can have measurable impacts on population health. Overall, this analysis demonstrates the usefulness of Bayesian methods because they offer reliable estimates and a clear approach to quantifying uncertainty, both of which are important for informing public policy and development planning.